

Generation Interconnection System Impact Study Report

For Queue Project Q24-2384

North Bluff 138 kV
October 27, 2025

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q24-2384. The Interconnection Customer proposes a 5.0 MW battery energy storage facility interconnecting at the North Bluff 138 kV substation in the Virginia area. The request was received on February 20, 2024. The study evaluated system intact and single contingency conditions using the regional planning case, including 367 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 6 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,170,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
North Bluff-Prince William Dp 138.0 kV Ckt 2	N Bluff-Prince William Dp Substation 138.0 kV Circuit 1	110.0	260	21.8
Twin Lakes Alte-T.S. Power 138.0 kV Line	Arcola-Twin Lakes Alte 138.0 kV	106.7	468	8.0
Mammoth Pacific 2-T.S. Power Substation 138.0 kV	Twin Lakes Alte-Arcola 138.0 kV	106.7	442	8.0
Mammoth Pacific 2-Wheeler 138.0 kV	Arcola-Twin Lakes Alte 138.0 kV	106.6	407	8.0
Chaffee Gas Recovery-S Jordan 138.0 kV Line	Wheeler-Pasco Cnty Solid Waste Resource Recovery 138.0 kV L	101.8	415	6.9
N Bluff-Prince William Dp 138.0 kV Line 1	Prince William Dp-N Bluff 138.0 kV	100.8	283	21.8

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
North Bluff-Prince William Dp 138.0 kV	Rebuild line with higher-capacity conductor	\$7,092,000	9% (\$638,000)
Twin Lakes Alte-T.S. Power 138.0 kV Line	Reconductor line	\$1,490,000	0% (\$3,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Mammoth Pacific 2 Substation-T.S. Power 138.0 kV	Reconductor line	\$1,406,000	0% (\$3,000)
Wheeler-Mammoth Pacific 2 138.0 kV	Reconductor line	\$2,203,000	0% (\$5,000)
S Jordan-Chaffee Gas Recovery 138.0 kV	Reconductor line	\$8,467,000	4% (\$334,000)
N Bluff-Prince William Dp 138.0 kV Line 1	Reconductor line	\$2,084,000	9% (\$187,000)
	Total allocated to project		\$1,170,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.